

Nick Hagerty

Updated June 2026

Montana State University
311 Linfield Hall
Bozeman, MT 59717

Email: nicholas.hagerty@montana.edu
Website: www.nickhagerty.com

Academic Positions

<i>Assistant Professor</i> Department of Agricultural Economics and Economics Montana State University	2020 – Present
<i>S.V. Ciriacy-Wantrup Postdoctoral Fellow</i> Department of Agricultural & Resource Economics University of California, Berkeley	2018 – 2020

Affiliations

<i>Invited Researcher</i> , J-PAL North America	2022 – Present
<i>Research Affiliate</i> , MIT Center for Energy & Environmental Policy Research	2021 – Present
<i>Invited Researcher</i> , King Climate Action Initiative (K-CAI)	2021 – Present
<i>Invited Researcher</i> , Agricultural Technology Adoption Initiative (ATAI)	2021 – Present

Education

Massachusetts Institute of Technology Ph.D., Economics. Dissertation: <i>Essays on the Economics of Water</i> Committee: Esther Duflo, Christopher Knittel, and Benjamin Olken	2018
Brown University Sc.B. <i>magna cum laude</i> , concentrations in Physics (with honors) & Economics	2010

Previous Positions

<i>Research Analyst</i> , NBER and J-PAL (Research Assistant to Esther Duflo, Michael Greenstone, and Rohini Pande)	2011 – 2012
<i>Research Assistant</i> , White House Council of Economic Advisers	2010 – 2011

Journal Articles (*published and conditionally accepted*)

["Transaction Costs and the Gains from Trade in Water Markets."](#) Conditionally accepted,
Review of Economics and Statistics.

- [“Externalities of Climate Adaptation in Common-Pool Groundwater Resources”](#) with Jeffrey Hadachek, Ellen Bruno, and Katrina Jessoe (2026). *Journal of Public Economics* 256: 105602. [\[preprint\]](#)
- [“Industrial Water Pollution and Agricultural Production in India”](#) with Anshuman Tiwari (2025). *Journal of the Association of Environmental and Resource Economists* 12(5): 1277–1315. [\[preprint\]](#)
- [“Anticipatory Effects of Regulating the Commons”](#) with Ellen Bruno (2025). *Journal of Environmental Economics and Management* 132: 103183. [\[preprint\]](#)
- [“The Tradeoff Between Wildlife Conservation and Renewable Energy: Evidence from Eagles and Wind Turbines”](#) with Brock O’Brien¹ (2025). *Land Economics*, 101(3): 285-303. [\[preprint\]](#)
- [“Market Power in California’s Water Market”](#) with Franceska Tomori, Erik Ansink, Harold Houba, and Charles Bos (2024). *American Journal of Agricultural Economics*. 106(3): 1274-1299.

Refereed Book Chapters

- [“The Political Economy of Groundwater Management: Descriptive Evidence from California”](#) with Ellen Bruno and Arthur Wardle (2023). In: Ariel Dinar and Gary Libecap, ed., *American Agriculture, Water Resources, and Climate Change*, National Bureau of Economic Research, University of Chicago Press.

Working Papers

- [“Adaptation to Surface Water Scarcity in Irrigated Agriculture.”](#)
(Reject and resubmit, *American Economic Review*.)
- [“Conservation Payments: Effectiveness and Design in Groundwater Irrigation”](#)
with Ariel Zucker.
- [“Does Micro-Irrigation Save Energy? An Investigation in Gujarat, India”](#)
with Piyush Gandhi and Ariel Zucker.
(Revise and resubmit, *Energy Policy*.)

Selected Work in Progress

- “Water Scarcity & Legal Conflict in Colorado” with Conner Harwood¹ and Philip Womble
- “The Benefits and Costs of Groundwater Regulation” with Ellen Bruno & Alvar Escriva-Bou
- “Do Property Rights Crowd Out Cooperation?” with Sungmin Cheu
- “A National Valuation of Irrigation Water”

¹ Former master’s students supervised at Montana State.

Self-Published Resources

California Surface Water Data (GitHub)	2022
Course Materials for “Advanced Data Analytics in Economics” (GitHub)	2022

Funded Grants

Foundation for Food and Agriculture Research (546,046 USD)	2026
Montana Water Center Faculty Seed Grant (24,987 USD)	2025
Individual and Multi-State Hatch Projects, USDA	2022 – Present
MSU Library Open Educational Resources (5,000 USD)	2022
International Growth Centre (19,254 GBP)	2022
Agricultural Technology Adoption Initiative (ATAI) (14,081 USD)	2021
King Climate Action Initiative (K-CAI) (7,560 USD)	2021
King Climate Action Initiative (K-CAI) (156,068 USD)	2020
International Growth Centre (45,000 GBP)	2020
Initiative for Regulation and Applied Economic Analysis, Montana State	2020
Weiss Family Program Fund	2019
International Growth Centre	2018
George and Obie Shultz Fund, MIT	2018
Abdul Latif Jameel Water and Food Systems Lab, MIT	2017
International Growth Centre	2017
Abdul Latif Jameel Water and Food Systems Lab, MIT	2016
Weiss Family Program Fund	2015
J-PAL Urban Services Initiative	2014

Fellowships & Awards

Lone Mountain Fellowship, Property & Environment Research Center	2025
S.V. Ciriacy-Wantrup Postdoctoral Fellowship, UC Berkeley	2018 – 2020
Tata Fellowship, MIT Tata Center for Technology & Design	2013 – 2015
Honorable Mention, NSF Graduate Research Fellowship Program	2013
MIT Energy Fellowship, MIT Energy Initiative	2012 – 2013
Honorable Mention, NSF Graduate Research Fellowship Program	2012
Mildred Widgoff Prize for Excellence in Thesis Preparation (Brown University Department of Physics)	2010

Conference and Seminar Presentations

- 2026: World Congress of Environmental and Resource Economists (scheduled)
- 2025: University of Chicago (Energy Policy Institute Junior Faculty Workshop), Association of Environmental and Resource Economists (AERE) Summer Meeting, Agricultural & Applied Economics Association (AAEA) Annual Meeting (Invited Paper), Property &

Environment Research Center (PERC), UC Berkeley (Agricultural & Resource Economics), Cornell (SC Johnson SEERE)

2024: UC Davis (Agricultural & Resource Economics), AERE Summer Meeting, Heartland Environmental and Resource Economics Workshop (UIUC)

2023: UC Santa Barbara (Bren School of Environmental Science and Management), AERE Summer Meeting, Montana State Applied Economics Summer Conference, Social Cost of Water Pollution Workshop (Cornell/EPA), North East Universities Development Conference (NEUDC), NBER conference on “Distributional Impacts of Climate Change in the Agricultural Sector”

2022: University of Maryland (Agricultural & Resource Economics), AERE Summer Meeting, Montana State Applied Economics Summer Conference

2021: Stockholm School of Economics (SITE)

2020: TWEEDS, University of British Columbia (CFREE/VSE), University of Hawai‘i Mānoa (Energy & Environment Seminar), NBER Summer Institute (EEE Workshop), AERE Summer Meeting (canceled), National Center for Environmental Economics (U.S. EPA), University of Massachusetts Amherst (Resource Economics), American Geophysical Union Fall Meeting

2019: Stanford (Environmental and Energy Policy Analysis Center), Occasional Workshop (UC Santa Barbara), Heartland Environmental and Resource Economics Workshop (UIUC), AAEA Annual Meeting, AERE Summer Meeting (Sponsored Session), Tinbergen Institute (Spatial Economics Seminar), Public Policy Institute of California

2018: Occasional Workshop (UC Santa Barbara), UC Berkeley (Environmental & Energy Economics Seminar), NBER Summer Institute (EEE Workshop), Interdisciplinary Ph.D. Workshop on Sustainable Development (Columbia), UC Davis (Agricultural & Resource Economics)

2017: Environmental Defense Fund

Courses Taught

Montana State University

ECNS 460/560: Advanced Data Analytics in Economics (master’s/undergraduate)

Fall 2021, Fall 2022, Fall 2023, Fall 2024, Spring 2026

ECNS 332: Economics of Natural Resources (undergraduate)

Fall 2020, Spring 2022, Spring 2023, Spring 2025, Spring 2026

AGBE 210: Economics of Agricultural Business (undergraduate)

Spring 2021

Massachusetts Institute of Technology

Kaufman Teaching Certificate Program

2016

14.41 Public Finance and Public Policy (undergraduate)

2015

(teaching assistant to Professor Jonathan Gruber)

Advising and Mentoring

Master's thesis advising (Placement) [*Outstanding thesis award]	
*Committee chair, 1 student (Research Associate, PERC)	2024 – 2025
*Committee chair, 1 student (Ph.D. program, UC Berkeley ARE)	2023 – 2024
Committee member, 1 student	2026 – Present
Committee member, 1 student	2024 – 2025
Committee member, 3 students	2022 – 2023
Committee member, 2 students	2021 – 2022
Summer mentor to PERC Graduate Fellow	2025
Undergraduate advising (approximately 20 students per semester)	2022 – Present
Co-supervision of full-time Research Associates (Placement)	
J-PAL South Asia (Research Manager, J-PAL South Asia)	2022 – 2023
Montana State (Economics Ph.D., Virginia Tech)	2021 – 2022
J-PAL South Asia (Policy Data Fellow, Government of Delhi, India)	2018 – 2020
J-PAL South Asia (Urban Studies and Planning Ph.D., MIT)	2017 – 2018
J-PAL South Asia (Urban Studies and Planning S.M., MIT)	2016 – 2017

Referee Service

Agricultural Economics
American Economic Journal: Applied Economics
American Economic Journal: Economic Policy
American Economic Journal: Macroeconomics
The American Economic Review
American Journal of Agricultural Economics
Applied Economics Teaching Resources
The Economic Journal
Environment and Development Economics
International Journal of Industrial Organization
Journal of the Association of Environmental & Resource Economists
Journal of Economic History
Journal of Environmental Economics and Management
Journal of the European Economic Association
Journal of Public Economics
Land Economics
Nature Sustainability
Review of Economics and Statistics
Review of Economic Studies
The Quarterly Journal of Economics
Water Resources and Economics

Professional Service

Co-founder and co-organizer, BIG SKY WORKSHOP on the Economics of Agriculture and the Environment (hosted by MSU and PERC)	2025 – Present
Member, Agriculture Land Valuation Advisory Council, State of Montana	2024 – Present
Grant reviewer, J-PAL North America	2025
Abstract reviewer, AERE Summer Meeting	2022 – 2025
Abstract reviewer, AAEA Annual Meeting	2022 – 2025
Grant reviewer, Digital Agricultural Innovations and Services Initiative	2022 – 2025
Grant reviewer, King Climate Action Initiative (K-CAI)	2021 – 2022
External reviewer, Environmental Defense Fund Discussion Paper Series	2021
Discussant, Standing Panel on Impact Assessment of the CGIAR	2021

Department Service

Organizer, DAEE Department Seminar & Brownbag Series	2025 – Present
Hatch Review Committee for Kelsey Larson	2025
Search committee, Assistant Professor of Agricultural Economics	2024 – 2025
Hatch Review Committee for Nicole Karwowski	2024
Graduate Affairs Committee	2021 – 2025
Committee to develop new major in Environmental Economics and Policy	2021 – 2023
Department Review Committee on Extension and Service	2021
Co-founder/organizer, Reading Group in Water Economics (UC Berkeley)	2018 – 2020

Outreach Publications

- [“How is SGMA affecting growers’ planting and drilling decisions?”](#) with Ellen Bruno (2025). *The Confluence*, June 24.
- [“California’s Sustainable Groundwater Management Act: What Are Its Early Impacts?”](#) with Ellen Bruno (2025). *ARE Update* 28(4): 2-4.
- [“Paying farmers for voluntary conservation can help solve the groundwater crisis”](#) with Ariel Zucker (2025). IGC Policy Brief IND-22167.
- [“Does industrial water pollution harm agricultural production?”](#) with Anshuman Tiwari (2025). *Ideas for India*, January 2.
- [“How farmers in India can be incentivised to save water”](#) with Ariel Zucker (2024). IGC Sustainable Growth Blog, June 5.
- [“Interior’s plan won’t solve the Colorado River crisis. Here’s what will.”](#) with Bryan Leonard (2022). *High Country News*, October 19.
- [“Energy efficiency in smallholder agriculture: Do micro-irrigation systems deliver?”](#) with Ariel Zucker (2019). IGC Policy Brief 89483.

[“Conservation credits: Payments for voluntary groundwater conservation”](#) with Ariel Zucker (2018). IGC Policy Brief 89456.

Articles for [AgEconMT.com](#), a publication for agricultural stakeholders in Montana

“The Economic Value of Irrigated Agriculture in Montana”	2026
Reprint: Trader’s Dispatch ; coverage: Northern Ag Network	
“A Quick Guide to Irrigated Agriculture in Montana”	2026
“Did a Robot Write This? The Future of Chatbots in Agriculture”	2023
“The value of clarifying water rights”	2022
“The Inflation Reduction Act: What’s in it for agriculture?”	2022
“The Inflation Reduction Act: What is it?”	2022
“Making people want to conserve water”	2022
“Recapping the 2021 drought”	2021
“The mystery of the missing drought impacts”	2021

Outreach, science communication, and policy commentary on social media 2017 – 2024
1.3 million total views on Twitter, 2020-2023.

Outreach Presentations

- “Transaction Costs and the Gains from Trade in Water Markets.” WestWater Research Lecture Series. Virtual, April 17, 2026.
- “Conservation Payments as Price Incentives: A Randomized Trial.” IWMI-Tata Water Policy Program Partners’ Meet. Anand, Gujarat, India (speaking remotely), December 4, 2025.
- “Irrigated Agriculture in Montana: Patterns, Challenges, and Opportunities.” Montana State Agricultural Economics Conference, Bozeman, MT, November 7, 2025.
- “Balancing Small-Scale Irrigation Solutions with Sustainable Groundwater Use.” ATAI and DAISI Research to Action for Resilient Agriculture Webinar Series. J-PAL and CEGA. Virtual, October 23, 2025.
- “Agricultural Impacts of Water Scarcity in the Short Term and Long Term.” Montana State Agricultural Economics Conference. Bozeman, MT, November 4, 2023.
- “Managing Water for Drought Resilience: An Economics Perspective.” Upper Missouri River Basin Drought and Human Health Workshop. Bozeman, MT, April 12, 2022.

Media Interviews and Appearances

Vox (“Let’s talk about the biggest cause of the West’s water crisis”)	2023
KUNC / NPR	2022
Montana PBS	2022
Volts newsletter by David Roberts	2021

Invited Contributions

“Arbitrage,” definition for the *Dictionary of Ecological Economics*

2021

Journal Articles in Physics

“[Statistics of DNA capture by a solid-state nanopore](#)” with Mirna Mihovilovic and Derek Stein (2013). *Physical Review Letters* 110, 028102.